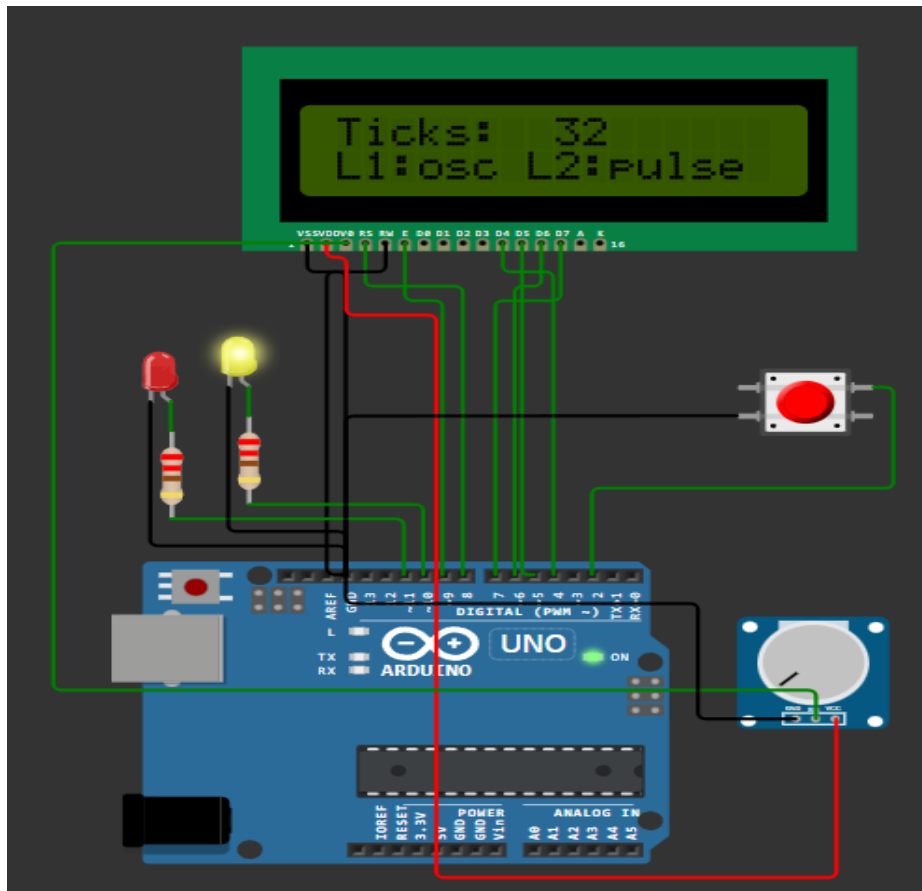


Question 3 — Periodic Tasks with the Timer Library



This project uses an Arduino Uno connected to a 16x2 LCD, two LEDs, a potentiometer, and a push button. The LCD works in 4-bit mode where RS and E are connected to pins 8 and 9, and data pins D4–D7 are connected to Arduino pins 4–7. The potentiometer is connected to the LCD V0 pin to control screen contrast. Two LEDs are connected to pins 10 and 11 through 220-ohm resistors, and their cathodes are connected to GND. The push button is connected to pin 2 and GND with the internal pull-up resistor enabled in the code. The program uses custom timer structures based on `millis()` to control different tasks without `delay()`. LED1 oscillates continuously, LED2 creates short pulses every 2 seconds, and the LCD shows the tick counter and LED status. When the button is pressed, the system enters a fast mode for 5 seconds where all timer speeds become faster.